S3 Supplementary QC and Robustness

Data-delivery validation, sample coverage, the R3 histogram corroboration check, and a reviewer-facing robustness summary

Article type: Supplement

Short title: S3 QC/Robustness - RichSleep_EAP_v2

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S3.0 Data-delivery validation (`01_validate_v3_delivery.py`)

Every check below passed (script exits 0; no `[FAIL]` lines in its console output):

- **`phase_resolved_v3`**: 24/24 subject files present; row counts match `run_meta_v3.json` for every subject; total 23,029 events. Bit-identity check against the companion articles' `event_phase_v2` on all 13 shared columns: identical (`subject`, `onset_sec`) key set and row order after sorting, and `np.allclose`/exact-string equality on every shared column - zero mismatched columns.
- **`slow_oscillation_v1`**: 24/24 files; covers both N2 (stage 2) and N3 (stage 3); all `so_qc_flag == "ok"`; 68,386 total (26,377 N2 + 42,009 N3).
- **`phase_continuous_v1`**: 24/24 files; sampling interval exactly 0.1 s (10 Hz) for all subjects; cardiac and respiratory phase-valid rate > 0.99 for all subjects; 8,834,120 total rows, mean 10.22 h/subject.
- **`event_phase_n3_so_v1`**: 24/24 files; total events (42,009) exactly matches `slow_oscillation_v1`'s N3 count; cardiac/respiratory phase-valid rate > 0.99 at both trough and upstate; `phase_coupling_qc_flag_trough` all `"ok"`.
- **`event_phase_rem_theta_v1`**: 24/24 files; total 2,412 epochs; `theta_phase_valid`/`cardiac_phase_valid`/`resp_phase_valid`/`autonomic_phase_eligible` all $= 1$ for every row; 1 subject has < 5 epochs (excluded from the per-subject-only arm, `S1.2`).
- **`non_event_risk_v1`**: 24/24 files; all rows `stage == 2` (N2) and `selection_status == "selected"`; cardiac and respiratory phase valid for all rows; 66,827 total, mean per-subject risk:spindle ratio consistent with the delivery's own ≈ 2.9 design target.
- **Cross-check**: all 190 companion-article geometry-valid events found in the v3 catalog by exact (subject, onset_sec) join; 25 SO-coupled across 15 subjects - confirms the plan's stated $N = 25$/15-subject power caveat exactly.

S3.1 Sample and coverage summary

Sample	N subj.	N events	Role
`phase_resolved_v3` full catalog	24	23,029	N2-spindle onset/peak phase-locking (R2), Test A/B cases (R3)
`event_phase_n3_so_v1`	24	42,009	N3-SO trough/upstate phase-locking (R2)
`event_phase_rem_theta_v1`	24 (23 for 2nd-order)	2,412	REM-theta phase-locking (R2)
`non_event_risk_v1`	24	66,827	Test A controls (R3)
`phase_continuous_v1`	24	8,834,120 rows	Test B surrogate source (R3)
Geometry-valid, N2-staged subsample (companion articles' population)	24	190	qe-interaction primary/robustness family (R4)
SO-coupled subset of the above	15	25 (21/19 after per-mode attrition)	SO-coupled descriptive check (R5)

Figure 1: Sample definitions and roles across R2-R5. Sources: `01_validate_v3_delivery.py` output tables and `02_build_merged_table.py`/`05_qe_interaction_moderation.py`.

The SO-coupled subset (25/190 = 13.2% of the geometry-valid sample) is far smaller than the primary qe-interaction family's 172/160 - this asymmetry is the reason R4 (well-powered, primary claim) and R5

(underpowered, descriptive-only) are reported with different evidentiary weight despite testing conceptually related questions.

S3.2 R3 histogram corroboration check (companion to the cross-null disagreement)

The main text (R3) and `04_spindle_onset_hazard_model.py`'s console commentary reference an "ad hoc 12-bin case-vs-control histogram comparison" used to judge which of Test A's (case-control) and Test B's (time-shift) disagreeing per-channel findings is more likely a genuine effect rather than a false positive of that specific null construction. This check is implemented in `06_make_figures.py`'s `fig3_hazard_curve` panel (Fig. `fig-hazard` in the main text): empirical case:control density ratio in 12 phase bins, overlaid with Test A's fitted $\exp(b_{\cos}\cos\varphi + b_{\sin}\sin\varphi)$ curve, one panel per channel.

Reading Fig. 3

Cardiac panel: the 12-bin empirical ratio scatters around 1.0 with no coherent monotonic trend matching the fitted curve's shape - visually consistent with Test A's own null result ($p = 0.379$) and with treating Test B's $p = 0.004$ hit as more likely a false positive of that specific surrogate construction than a real effect. **Respiratory panel**: the empirical ratio shows a small but visually coherent, monotonic excess/deficit pattern tracking the fitted first-harmonic curve - consistent with Test A's highly significant ($p = 3.96 \times 10^{-7}$) but small effect, and with Null 0′s directionally-consistent (though non-significant) raw $p = 0.0265$. This visual check is descriptive, not a separate formal statistical test; it is used only to adjudicate which of two already-computed, disagreeing p-values is more consistent with a visible shape in the data, not to generate a new p-value of its own.

S3.3 R4/R5 robustness detail

The primary qe-interaction family (`S1.5`, `S2.4`) was repeated with three operationalizations of intrinsic spindle strength (log1p sigma-power z-score, log1p amplitude, untransformed sigma-power z-score); none produced any BH-FDR-significant interaction term (minimum FDR q: 0.834, 0.988, 0.725 respectively, across the three families). No single (delta_mode, endpoint) pair approaches significance under more than one operationalization simultaneously - there is no cross-transform-convergent candidate interaction being missed by the correction, only three independently null families.

The main-effect (non-interaction) `resp_peak_alignment` coefficients in this interaction model are nominally in the same direction as the companion article's own main-effect finding for the two endpoints it flagged as fragile BH-FDR survivors (`d`: this model's $p = 0.023$ vs. companion's $q = 0.021$; `d_l`: $p = 0.0030$ vs. companion's $q = 0.005$, both `event_prefar` mode) - but this model's main-effect p-value is not itself BH-FDR corrected within this family (the **interaction** term is the pre-specified quantity of interest here), so this consistency is reported as a qualitative cross-check, not as independent statistical confirmation of the companion article's finding.

S3.4 Reviewer-facing robustness summary

Robustness summary

Cross-carrier sweep (R2, primary claim of this article). N3-SO phase-locking (trough and upstate, both channels) is significant under every statistic tried (pooled permutation R_1 and T_2, second-order Rayleigh, per-subject Rayleigh survivor rate 16-17/24) and is the strongest effect in the dataset by a wide margin. N2-spindle-peak reproduces the companion article's own numbers exactly (regression check, R2). REM-theta is uniformly null (all $p > 0.32$, 0-1/24 per-subject FDR survivors). This ordering is treated as a robust, internally consistent result. **Onset-hazard tests (R3).** Three independent, individually defensible null constructions disagree in different directions per channel; this disagreement, and the histogram check used to adjudicate it (`S3.2`), is the reported result - not resolved to a single number. **qe-interaction (R4-R5).** A clean null across three qe operationalizations (54 tests total) and an underpowered, non-confirmatory descriptive check on the small SO-coupled subset. Open items not yet closed: the SO-coupling power gap (would need a substantially larger SO-coupled-event yield to become a well-powered test), the single-dataset limitation (no ANPHY/BOAS phase-resolved N3-SO/REM-theta export exists), the absence of any geometry outcome for N3-SO/REM-theta (so R2′s carrier ordering cannot itself be

linked to a downstream functional consequence the way R4 attempts for spindles), and the histogram check's own descriptive (non-formal-test) status.

Bibliography